

CANDIDATE DIGITAL BIOMARKERS OF MASKING IN NEURODIVERGENT ADULTS

Phase II Feasibility Assessment

First cohort: adult women of West African heritage, resident in Berlin

Index condition: ADHD. Masking studied across ADHD, autism and co-occurring presentations

Proposed primary site: Charité Universitätsmedizin Berlin

"The signal is present.

The instrument to receive it has not been built."

Status: working draft. This is a proposed study. No site or partner has agreed to participate. No candidate measure described here is a validated biomarker of masking, and nothing in this document is a diagnostic claim.

Contents

01 — The Problem This Study Is Addressing	3
02 — Why the Measurement Problem Is Real	3
What Masking Is, and What It Is Not	3
Three Proposed Mechanisms, One Clinical Surface	4
03 — Why Berlin. Why These Sites. Why Now.	5
04 — The Candidate Biomarker Register	5
Register of Candidates	6
Tier One — Established Signal: Actigraphy and Rest-Activity Rhythm	6
Tier Two — Established Adjunct: Continuous Performance Testing	7
Tier Three — Pilot Stage: Smartphone Passive Monitoring	7
Tier Four — Design Stage: Context-Labelled Self-Report	7
Hypothesis-Stage Candidates and the Masking Load Index	8
05 — The First Cohort: Design and Culture Calibration	8
06 — Site Profile	9
07 — Risk Matrix and Mitigation Strategy	9
08 — Regulatory and Data Governance Framework	10
09 — What This Study Proposes to Demonstrate	11
10 — Forward Signal	12
11 — References	12

01 — The Problem This Study Is Addressing

The signal is present. The instrument to receive it has not been built.

To our knowledge, no digital biomarker for adult ADHD has been qualified by the European Medicines Agency, and none exists for masking in any neurodivergent condition. The qualification framework exists. The qualified biomarker does not. What exists instead is a diagnostic infrastructure calibrated to performance: to the clinical presentation of inattention, impulsivity and dysregulation in the room where assessment happens. That infrastructure has a structural blind spot. The hypothesis this study is built on is that the population most consistently falling through it is not the most severely affected. It is the most adapted.

This feasibility study proposes to begin building the instrument, for masking: the sustained suppression of natural response to meet environmental demands, and the cost of doing so. Masking is studied here across neurodivergent presentations, with ADHD as the index condition and autism and co-occurring presentations in scope. The first cohort is chosen because its diagnostic invisibility is structural rather than incidental: adult women of West African heritage resident in Berlin. The cohort and the instruments are culture-calibrated from the start, not adapted afterwards.

The study combines actigraphy, continuous performance testing, smartphone passive monitoring and context-labelled self-report across a seven-day window. The feasibility question is not whether these candidates are biomarkers of masking. It is whether they can be collected, from this population, with enough completeness, retention and acceptability to justify a validation study.

That is what this document proposes. What it is certain about, what it is interpreting, and what it is genuinely still finding out are all named here, and kept apart.

02 — Why the Measurement Problem Is Real

Adult ADHD and autism in high-functioning populations, and in women of West African heritage specifically, may not present to assessment the way assessment was designed to receive it. Three mechanisms are proposed. Each produces the same composed clinical surface. Current diagnostic tools distinguish between none of them. They are hypotheses, informed by lived experience and adjacent literature, and each carries a claim-status note.

What Masking Is, and What It Is Not

Masking, as used here, is the deliberate or automatic suppression of natural response to meet environmental demands. Not deception. Not inauthenticity. Adaptation, with a cost. In autism research the construct is studied as camouflaging (Hull et al., 2017) and has a validated self-report measure, the CAT-Q (Hull et al., 2019). In ADHD the construct is less developed and, to our knowledge, has no widely validated masking measure.

Masking overlaps with neighbouring constructs that this study must separate from it: code-switching, acculturative and minority stress, and culturally shaped norms of strength and composure, such as the Superwoman Schema described in African American women (Woods-Giscombé, 2010). This is why a matched comparison group of women of the same heritage without neurodivergence is a design requirement. Without it, a cost signal cannot be attributed to neurodivergent masking rather than to shared stressors.

Three Proposed Mechanisms, One Clinical Surface

Mechanism One — Physiological Adaptation

A neurodivergent person managing chronic sensory and cognitive load over years may develop a shifted baseline. What registers as severe impairment in a body without that history may register differently in one that has been running the load for decades. The pain literature documents a parallel problem in Black patients: clinicians systematically underestimate reported pain, and false beliefs about biological difference predict that underestimation (Hoffman et al., 2016). Whether acquired adaptation, rather than assumed biological difference, is the better explanation is the hypothesis this study takes from that literature. It is not a finding of it. If the same attribution error operates in attention assessment, the scale was not calibrated on this body.

Claim status: Hypothesis. Adjacent evidence: clinician underestimation of pain in Black patients (US data); the cumulative physiological cost of chronic stress (allostatic load; McEwen, 1998; Geronimus et al., 2006). Gap: none of this has been tested in neurodivergent women, or in women of West African heritage in Europe.

Mechanism Two — Conditioned Suppression

In many West African households, and across the diaspora they produced, visible expression of difficulty was not available as a behaviour. Composure was survival. By the time a woman shaped by this enters a diagnostic room, the hold may no longer be a strategic choice. It may be how the nervous system operates. The assessment receives a performance it cannot read as data.

Claim status: Interpretation from lived experience; not yet documented in the literature for this population. West Africa is not one culture: households, ethnic groups, faiths and migration histories differ. Composure is treated here as a valued, adaptive norm whose cost has gone unmeasured, not as a deficit in the culture. The formative phase (section 05) tests where this holds and where it does not.

Mechanism Three — Absent Infrastructure

Many women from healthcare systems where access was contingent on payment, where self-advocacy was not the operating assumption, where you did not perform for a right you were never taught you had, arrive in European clinical settings without the grammar of entitlement. The assessment reads the silence as absence of symptom. The silence can be the absence of a tool that was never in the kit.

Claim status: Interpretation, consistent with documented barriers to care for migrant women; no neurodivergence-specific data for this population. The cohort will include first-generation and German-born women. Mechanism Three is expected to apply differently to each, which is testable.

Three proposed mechanisms. One clinical surface. One interpretation. Possibly wrong three ways at once.

Passive monitoring is proposed for one shared reason: it does not depend on the performance the clinical room expects. Actigraphy and smartphone interaction data are collected where the masking load is actually running, not in the appointment, where the composed surface is exactly what gets measured. Continuous performance testing is kept on purpose, as the performance surface: the thing the clinical room sees. The candidate signal is the gap between the two. Passive does not mean unobserved. Participants know they are monitored, and reactivity is a named limitation (section 07).

The working hypothesis is that a woman managing sustained masking across a working week, filtering communication output, regulating sensory environment, sustaining social performance within tolerable limits, is not resting between appointments, and that the cost accumulates: by midweek, disrupted sleep, elevated baseline fatigue, progressive depletion of the reserves that make the surface possible. She continues. The seven-day window can test this prediction directly, by comparing work days with non-work days within the same woman.

The somehow she continues is the data. Whether passive monitoring can measure it is the question.

03 — Why Berlin. Why These Sites. Why Now.

Berlin is proposed as the location for this study because the population is here and the infrastructure to recognise it is, in places, not.

The city has a significant and growing West African immigrant community. The eligible subgroup, women of West African heritage with suspected or confirmed neurodivergence, has never been counted, and counting it is one of the outputs of this study. The city has at least one academic centre with documented adult ADHD diagnostic capacity and documented demand exceeding that capacity. It has a healthcare system that is, in specific pockets, developing the reflex to recognise neurodivergence in non-standard presentations, and in others has not yet begun.

Lived-experience accounts suggest that recognition of neurodivergence in women of West African heritage is uneven in both the United Kingdom and Germany, and that Germany has built less of a reflex for it. This is an observation, not a finding. It is not relied on in the design, and a formal comparison is listed as a gap (section 10).

Site Selection Rationale

Charité Universitätsmedizin Berlin is the proposed primary research site. The rationale is operational: documented infrastructure for ADHD assessment, a separate autism outpatient service, academic research capacity, and closed registration that constitutes direct evidence of unmet diagnostic demand. Charité is proposed not because it is perfect but because it has the structural capacity to run this study and documented proof that the population it would serve is already attempting to access it. No agreement with Charité exists at the time of writing.

Closed registration is not a recruitment obstacle. It is part of the recruitment argument. It is also the most ethically sensitive feature of the design. Offering a reference-standard assessment to women who cannot access one risks undue inducement and the appearance of queue-jumping. The safeguards: the assessment is a study procedure, not a route around the clinic; results are returned to the participant and, with her consent, to her clinician; consent materials state plainly that a research assessment is not a guarantee of clinical diagnosis or treatment; and the inducement question goes to the ethics committee before any recruitment.

A second Berlin site is to be identified against the same criteria: documented ADHD and autism diagnostic service, research capacity, ethics pathway, language access. A study that cannot articulate the difference between a functional and a non-functional site has a preference, not a feasibility framework.

04 — The Candidate Biomarker Register

This section treats every measure as a candidate, not a biomarker. Terms follow the FDA-NIH BEST framework: a biomarker is defined by what it indicates and the context of use it is intended for, and qualification is a statement about that pair. For each candidate the register states what it is hypothesised to indicate, how strong the evidence is for ADHD and for masking separately, what else could produce the signal, and what calibration the first cohort requires. No candidate is presented at a higher evidence level than the literature supports. None currently has direct evidence as a biomarker of masking.

Intended context of use for this phase: exploratory and research-only; candidate monitoring and risk-stratification signals of masking load in adults with suspected or confirmed neurodivergence. Not diagnostic. Not for clinical decisions.

† marks a figure or statement carried forward from version 1 and not independently re-verified in this revision. Check against the source before external circulation.

Register of Candidates

Candidate	Hypothesised to indicate	Evidence: ADHD and masking	Main confounders	Calibration for cohort 1
C1. Rest-activity rhythm (wrist actigraphy)	Sleep-circadian cost of sustained effort; failure to recover across the week	ADHD: emerging in adults (Walter 2026; Tonetti 2018; Boonstra 2007). Masking: none direct.	Shift and night work, caregiving, depression, anxiety, sleep apnoea, hormonal transitions, medication	Accelerometer-only device; work-hours log; start day set so the window includes non-work days
C2. Continuous performance test (QbTest+ or equivalent)	The performance surface the clinical room sees	ADHD: established adjunct, limited discriminative validity against clinical comparators. Masking: not a masking measure; used as the comparator.	Anxiety in the test setting, practice effects, language of instructions, fatigue	Manufacturer norms not assumed to fit; the within-study comparison group is the primary reference
C3. Smartphone passive monitoring (notification latency, light variability, session duration)	Lapses and regulatory load in daily life	ADHD: pilot only (20 v 20, ages 16 to 39). Masking: none.	Phone rules at work, shared phones, battery and data limits, operating-system differences, shift work	Work-context labels; local data processing; separate consent for each data stream
C4. Context-labelled self-report (short daily check-in; baseline masking scale)	Perceived masking effort by context; within-person contrast between high- and low-masking settings	ADHD: no widely validated masking measure to our knowledge. Masking: CAT-Q validated in autistic adults only.	Response style, stigma, social desirability, translation drift	Vocabulary and items co-developed in Phase 0; forward-back translation; cognitive interviewing
C5. Autonomic signals (heart-rate variability, electrodermal activity)	Autonomic cost of sustained regulation	ADHD: heterogeneous, not established. Masking: none.	Posture, caffeine, exercise, anxiety; optical sensors can be less accurate across skin tones	Not in cohort 1 core. If added: ECG-based reference and signal-quality checks
C6. Neuroendocrine (salivary diurnal cortisol; hair cortisol)	Cumulative stress-axis load (allostatic load)	ADHD: not established. Masking: none; rationale from the allostatic-load literature.	Sampling burden; hair texture and chemical treatment affect hair cortisol; shift work	Not in cohort 1. Revisit after Phase 0 with the advisory panel
C7. Masking Load Index (composite)	Divergence between the performance surface and the cost signals	Construct only. No evidence for ADHD or for masking.	Inherits every confounder above	Defined a priori; components collected in cohort 1; the composite itself is not evaluated

Tier One — Established Signal: Actigraphy and Rest-Activity Rhythm

Protocol: Seven-day continuous wrist actigraphy on an accelerometer-only device, including at least two non-work days. Outcome variables: interdaily stability (IS), intradaily variability (IV), L5 onset timing, relative amplitude, sleep efficiency, and the within-person difference between work days and non-work days.

Walter et al. (BMC Psychiatry, 2026) provide the primary adult evidence base: 54 adults with ADHD monitored over 10 days against 47 controls showed significantly lower IS in the ADHD group ($M = 0.450$ versus $M = 0.488$, $p = 0.029$, moderate effect size)[†], with delayed L5 onset that is consistent with a circadian phase shift. Tonetti et al. (Journal of Psychiatric Research, 2018) identified absence of a post-lunch dip in motor activity as a potential trait marker of adult ADHD[†]. Boonstra et al. (Sleep, 2007) used a seven-day window for rest-activity characterisation in adult ADHD[†], and Coogan and McGowan (2017) review circadian function in ADHD.

The adult evidence base remains thin relative to the paediatric literature. That gap is part of this study's scientific justification: the signal exists and is measurable in adults, but adult-specific validation is what is absent. None of these studies was designed to measure masking, and, to our knowledge, none enrolled women of West African heritage as a defined group.

Evidence level: Established signal for ADHD, adult validation emerging; not established for masking. | Tier: Primary.

Tier Two — Established Adjunct: Continuous Performance Testing

Protocol: Single 20-minute CPT session at baseline, administered in-site. Outcome variables: omission errors (inattention), commission errors (impulsivity), reaction-time variability.

Role in this study: the performance surface. A CPT is included because it is what the clinical room sees. The hypothesis is that some masked participants will perform within the normative range while showing high cost on the rest-activity and smartphone measures. A dissociation between performance and cost, not the CPT score itself, is the signal of interest.

Edebol et al. (PsyCh Journal, 2013) report sensitivity of 86% and specificity of 83% for a composite QbTest+ measure when comparing ADHD patients with healthy controls†. The figure weakens against clinical comparators. In a specialist outpatient sample (94 adults with ADHD and 20 other psychiatric patients), Brunkhorst-Kanaan et al. (Frontiers in Psychiatry, 2020) found that the most discriminating QbTest parameter, QbActivity, reached sensitivity of 68% and specificity of 48% at the recommended cutoff (AUC 0.65), and concluded that CPTs did not help identify ADHD in that setting. The clinical utility of QbTest+ therefore lies in supporting comprehensive assessment rather than replacing it.

Charité's diagnostic methodology is reported to include neuropsychological procedures, including continuous performance testing†. This study is not introducing a foreign instrument. It is adding digital biomarker candidates to an existing diagnostic pathway. Manufacturer norms are not assumed to fit this population.

Evidence level: Established clinical adjunct; limited discriminative validity against clinical comparators. | Tier: Primary comparator.

Tier Three — Pilot Stage: Smartphone Passive Monitoring

Protocol: Seven-day passive monitoring via the RADAR-base platform or equivalent, hosted locally. Outcome variables: notification response latency (questionnaire and social or communication applications), ambient light variability, active session duration.

Sankesara et al. (JMIR Formative Research, 2025) provide the foundational prospective observational evidence: 20 people with ADHD (ages 16 to 39, adults and adolescents) and 20 comparison participants were monitored for 10 weeks using the ADHD Remote Technology system, built on RADAR-base. Five of ten digital signals showed significant group differences with moderate to large effect sizes ($d = 0.64$ to 1.13 †). Notification response latency and ambient light variability were among the strongest signals. The authors call for larger studies to assess whether these markers can track severity or predict outcomes. This feasibility study sits in exactly that pipeline position.

The seven-day window here is shorter than the 10 weeks of that study. This is a deliberate feasibility decision: participant burden is a plausible retention risk in this population, and the shorter window tests whether meaningful signal can be obtained within a practically sustainable timeframe. Notification latency is also sensitive to workplace phone rules and hands-on shift work, which is why every day carries a context label.

Evidence level: Prospective observational, pilot stage, large effect sizes, explicit call for replication. | Tier: Exploratory.

Tier Four — Design Stage: Context-Labelled Self-Report

Protocol: A check-in of about two minutes each day, in the participant's strongest language: where she was, with whom, which language she spoke, and how much effort it took to appear the way she wished to appear. A baseline masking scale (the CAT-Q, used as a reference and not as a validated instrument for this population) and the site's structured reference assessment for diagnostic status complete the picture.

This tier exists because masking is a within-person, context-dependent phenomenon. The same woman in a ward, a church, a family kitchen and a clinic room is a natural experiment. Context labels let the passive signals be read against the settings that produce them.

Evidence level: Design stage. The masking scale is validated in autistic adults only; no validation in ADHD or in West African-heritage samples. | Tier: Exploratory, co-designed.

Hypothesis-Stage Candidates and the Masking Load Index

Autonomic signals (C5) and neuroendocrine measures (C6) are deliberately outside cohort 1. The evidence is weakest for them, the participant burden is highest, and two of the methods carry known calibration problems in this population (optical heart-rate sensors across skin tones; hair-based sampling across hair textures and treatments). They return to the table only after the formative phase and with the advisory panel's agreement.

The Masking Load Index (C7) is the proposed composite: the divergence between the performance surface and the cost signals across Tiers One to Four. It is a construct, not yet a measure. Cohort 1 tests whether its components can be collected. Whether the composite means anything is a question for a later study.

05 — The First Cohort: Design and Culture Calibration

Who the First Cohort Is

Group A: adult women resident in the Berlin-Brandenburg region, with self-identified West African heritage, who have confirmed or suspected ADHD and/or autism and complete a structured reference assessment as part of the study. **Group B:** a comparison group of women of the same heritage without neurodivergence, matched on age band and working pattern, especially shift work. Heritage is self-reported, never inferred. The proposed working definition is self or at least one parent born in a West African country, to be finalised with the community advisory panel. First-generation and German-born women are recorded separately. Indicative target: 30 to 40 participants across both groups, to be confirmed by a biostatistician.

What the First Cohort Can and Cannot Show

It can show whether these candidates can be collected from this population: recruitment, retention, wear-time, completeness, acceptability, data quality, and a first descriptive look at signal. It cannot show that any candidate is a biomarker of masking. It cannot establish cultural specificity, because it has no non-West-African comparison group; that needs a second cohort and is named here as a dependency, not assumed. It cannot be generalised to all West African-heritage women in Germany, or to women in West Africa. It is not powered to separate ADHD from autism; condition is a pre-specified descriptive stratum.

Culture Calibration Is Built In, Not Bolted On

Phase 0, formative work before any data collection. Roughly 8 to 12 interviews and one or two focus groups with women of West African heritage who have lived experience of late or missed diagnosis, plus clinicians. Outputs: the vocabulary women actually use for neurodivergent experience before diagnosis, acceptability of devices and burden, the language of consent, and review of every candidate measure.

A paid community advisory panel. Women of West African heritage and community organisations review the protocol, materials and return-of-results policy, and can flag harms.

Language and translation. Consent and instruments in the participant's strongest language, with interpreters where needed. Forward-back translation and cognitive interviewing for any instrument before use. The CAT-Q is treated as unvalidated in this population.

Device and sample equity. Accelerometer-only actigraphy is the core measure. Optical heart-rate sensors are excluded from primary analysis because accuracy differences across skin tones are documented (Colvonen, 2020). No hair-based sampling in cohort 1.

Real lives as design constraints. Shift work, night work, caregiving and multiple jobs are recorded in a work-hours log, and each day is labelled by context. The monitoring window is scheduled to include non-work days.

Norms. Manufacturer CPT norms are not assumed to apply. The within-study comparison group is the primary reference.

Researcher positionality. The lead investigator shares heritage with the target population. This is treated as an asset for trust and a source of potential bias, mitigated by a pre-specified analysis plan and independent

statistical review.

Return of results and benefit. Participants can opt in to an individual summary. It is a research result, not a diagnosis. With consent, it can be shared with their clinician.

Pre-Specified Feasibility Endpoints

Endpoint	What is counted	Indicative progression criterion
Recruitment	Eligible women approached who enrol	Target set after Phase 0 estimates the eligible pool
Retention	Enrolled women who complete the seven-day window	At least 80%
Wear-time	Participants with at least 5 valid actigraphy days, including at least 1 non-work day	At least 70%
Daily check-ins	Check-ins completed	At least 70%
Acceptability	Structured feedback on burden, privacy and language	Qualitative themes; no participant-reported harm
Data quality	Usable data per device stream	Reported per stream; no pooling before review

Indicative values are proposals for discussion. They are not evidence-based thresholds and must be fixed before recruitment with a biostatistician.

06 — Site Profile

Proposed Primary Site: Charité Universitätsmedizin Berlin

Element	Charité Universitätsmedizin Berlin	Verification status
ADHD diagnostic service	ADHS-Sprechstunde, Campus Mitte, documented and active	Confirmed via a third-party listing; re-verify on the Charité site
Current demand status	Not accepting new registrations “due to very high demand”	Confirmed via a third-party listing; page undated; re-verify
Autism service	Separate Ambulanz für Autismus-Spektrum-Störung	Listed on the Charité site; scope to be confirmed
CPT infrastructure	Neuropsychological procedures including CPT reported in diagnostic methodology	† Not re-verified
Academic research capacity	Full university hospital: ethics committee, research coordinators, IMP management	General knowledge; confirm with the site
Ethics committee	Approximately 8 weeks to Votum	† Not re-verified
Referral pathway	Facharztüberweisung; Berlin/Brandenburg catchment	† Not re-verified
Site designation	PROPOSED PRIMARY RESEARCH SITE	No agreement exists at the time of writing

A second Berlin site is to be identified against the criteria in section 03. No other institution is named in this version.

07 — Risk Matrix and Mitigation Strategy

Category	Specific risk	Prob.	Impact	Mitigation strategy
Construct validity	Signal reflects shared stressors (minority and migration stress, shift work), not neurodivergent masking	High	High	Matched comparison group; work-pattern matching; context labels; claims limited to feasibility
Reference standard	“Suspected” cases without confirmed diagnosis leave case status uncertain	High	High	Structured reference assessment in the study; confirmed and suspected analysed separately; no accuracy claims

Category	Specific risk	Prob.	Impact	Mitigation strategy
Recruitment	Closed registration at the proposed site, no agreement yet, small eligible pool	Med-High	High	Phase 0 estimates the pool; community organisation and GP network outreach; second site; no dependency on the waitlist alone
Retention	Study burden given masking load; privacy concerns about phone monitoring	Med-High	High	Short window; passive collection; compensation; flexible scheduling; panel input on burden
Confounding	Shift work, caregiving, sleep disorders, hormonal state, medication	High	Medium	Work-hours log; covariates recorded; stratified descriptive analysis
Measurement equity	Device and instrument bias across skin tone; instruments unvalidated in this population	Medium	High	Accelerometer-only core; optical sensors exploratory; translation and cognitive interviewing; internal comparator
Ethics	Diagnostic access as incentive may be judged undue inducement	Medium	High	Ethics review first; assessment as a study procedure; explicit consent language; no clinic-pathway advantage
Regulatory	Study status unclear (clinical investigation or research use of instruments); later diagnostic claims would fall under MDR Rule 11	Medium	Medium	Early consultation with the ethics committee and, if needed, BfArM; research-only use; no diagnostic claims in cohort 1
Data governance	GDPR Article 9 health data and self-reported heritage (special category)	High	High	DPIA before recruitment; pseudonymisation at source; local processing; granular consent with partial withdrawal
Generalisation	Berlin cohort read as representative of Sierra Leone or West Africa	Medium	High	Explicit scope statement: methods transfer, results do not
Reactivity	Being monitored changes behaviour	Medium	Medium	Named limitation; sensitivity analysis excluding the first monitoring day
Biomarker signal	Signal-to-noise of novel candidates insufficient	Medium	Medium	Prior effect sizes inform planning; exploratory designation; success criteria set a priori
Composite performance	Unknown interaction between modalities	Unknown	High	Pre-specified hierarchy of primary and exploratory measures; composite not evaluated in cohort 1

08 — Regulatory and Data Governance Framework

Two Pathways, Two Questions

EMA qualification of a novel methodology concerns whether a biomarker or endpoint can be relied on in medicine development, within a stated context of use. EU MDR concerns whether software intended to inform diagnosis may be placed on the market. They are separate routes with separate evidence requirements. A diagnostic aid runs through MDR. An endpoint for drug trials runs through EMA qualification. Cohort 1 is on neither route. It is a feasibility study that decides which questions are worth asking.

EU MDR Classification

A composite battery used as software-informed clinical decision support would fall under EU MDR 2017/745 as Software as a Medical Device. Under Rule 11, software intended to provide information used to take decisions with diagnostic or therapeutic purposes is classified Class IIa at minimum, and higher where decisions may cause serious deterioration in health. Class IIa and above requires a Notified Body, which assesses the manufacturer's conformity, not a research protocol. The battery cannot be self-certified.

Cohort 1 makes no diagnostic claims. Whether it counts as a clinical investigation under MDR Article 62 and the German MPDG, or as a research study that uses instruments as measurement tools, is to be settled with the ethics committee and, if needed, BfArM before the protocol is finalised. The CPT is used within its CE-marked intended purpose†; the actigraphy device is a research data-collection tool; the smartphone platform is research-only.

Regulatory anchor: EU MDR 2017/745, Annex VIII Rule 11. EMA Qualification of Novel Methodologies guidance, EMA/CHMP/SAWP/72894/2008 (check current revision).

GDPR and Data Protection

Continuous behavioural monitoring of neurodivergent adults is high-risk processing of special category health data under GDPR Article 9, and self-reported heritage is itself special category data (racial or ethnic origin). The proposed legal basis is explicit consent (Article 9(2)(a)) together with scientific research provisions (Article 9(2)(j) and Article 89), with German supplementary rules (BDSG sections 22 and 27) and Berlin hospital-law research provisions to be confirmed with the site's data protection officer. A Data Protection Impact Assessment is mandatory before recruitment. The privacy architecture is a design condition, not a compliance addendum.

Communities that have experienced research as extraction have good reason to ask who benefits from their data. The consent architecture therefore includes: explicit, granular, informed consent in the participant's strongest language; the right to withdraw individual data streams without withdrawing from the study; data held in the EU and not reused without new consent; clear retention and destruction timelines; participants able to see what was collected about them; and a plain-language privacy notice that does not assume familiarity with European data protection frameworks.

Regulatory anchor: GDPR Articles 9 and 89; BDSG sections 22 and 27; ICH E6(R3), whose principles and Annex 1 have applied in the EU since 23 July 2025.

09 — What This Study Proposes to Demonstrate

This study is built on a problem worth studying and a design that can be defended. The uncertainty it names is genuine and methodologically honest. Claims are sorted by how much weight they can bear.

What Is Established

Rest-activity differences separate adults with ADHD from controls at group level, with adult evidence still emerging. CPT has limited discriminative validity against clinical comparators. Clinicians systematically underestimate pain in Black patients, linked to false beliefs about biological difference. Camouflaging is measurable by self-report in autistic adults. To our knowledge, no qualified digital biomarker of masking exists.

What Is Supported by Adjacent Evidence

Smartphone passive signals separated ADHD from comparison participants in one pilot. Chronic stress carries cumulative physiological cost. Demand for adult ADHD assessment exceeds capacity at at least one Berlin academic centre.

What Is Hypothesised

The three mechanisms in section 02. That masking cost accumulates across a working week and can be seen in rest-activity and smartphone signals. That the gap between performance and cost is a usable signal. That all of this applies to women of West African heritage in Berlin, and differs between first-generation and German-born women.

What Is Genuinely Open

Whether seven-day passive monitoring produces data of sufficient signal-to-noise in this population to justify a validation study. Whether participants can be recruited and retained in the numbers needed. Whether signals are specific to neurodivergent masking or shared with other stressors. Whether a composite outperforms any single modality, and by how much. Whether any of it transfers to West African settings.

These are the feasibility questions. They are open because this study has not been done. Claiming otherwise would not be science. It would be the same attribution error this study is designed to correct.

The aim is not to prove what we already suspect. It is to find out whether the instrument that makes it measurable can be built: reliably enough that a regulator could one day trust the measurement, and specifically enough that the population it was built for is no longer invisible inside it.

10 — Forward Signal

This feasibility assessment is the front end of a larger protocol concept currently in development. The Phase II questions addressed here, specifically site viability, retention, and signal quality across the candidate tiers, generate the data needed to decide whether a pivotal validation study is justified, and to power it if so. That protocol exists as a working document and will be made available to the right institutional partner at the appropriate stage of engagement.

To our knowledge, no systematic review of digital biomarker validation in adult ADHD in European populations, and none addressing masking, exists in the peer-reviewed literature. The evidence mapping in section 04 is a rapid overview, not a systematic review. A formal systematic or scoping review is a planned output of the research trajectory this document initiates.

Beyond Berlin

The method travels. The results do not. What cohort 1 produces for West African settings is a protocol for culture calibration: how to build a vocabulary, a consent process, an instrument battery and a governance model with the women a diagnostic system has not seen. Nothing here collects data in Sierra Leone or any other West African country. Any work there would need its own ethics approval, local partners and a statutory institutional home, and would begin from locally defined categories rather than imported ones.

11 — References

Biomarker framework

FDA-NIH Biomarker Working Group. BEST (Biomarkers, EndpointS, and other Tools) Resource. Silver Spring (MD): FDA; Bethesda (MD): NIH; 2016.

Masking and camouflaging: qualitative account

Hull L, Petrides KV, Allison C, et al. "Putting on my best normal": social camouflaging in adults with autism spectrum conditions. *J Autism Dev Disord*. 2017;47(8):2519–2534.

Masking and camouflaging: measure

Hull L, Mandy W, Lai MC, et al. Development and validation of the Camouflaging Autistic Traits Questionnaire (CAT-Q). *J Autism Dev Disord*. 2019;49(3):819–833.

Tier One: actigraphy

Walter A, Martz É, Weiner L, et al. Sleep, rest-activity rhythm, cognitive and emotional symptoms in adult ADHD: unraveling the links with an actimetry-based approach. *BMC Psychiatry*. 2026;26:317. <https://doi.org/10.1186/s12888-026-07947-9>

Tier One: circadian trait marker

Tonetti L, Conca A, Giupponi G, Filardi M, Natale V. Circadian activity rhythm in adult attention-deficit hyperactivity disorder. *J Psychiatr Res*. 2018;102:254–260.

Tier One: adult actigraphy baseline

Boonstra AM, Kooij JJS, Oosterlaan J, Sergeant JA, Buitelaar JK, Van Someren EJW. Hyperactive night and day? Actigraphy studies in adult ADHD: a baseline comparison and the effect of methylphenidate. *Sleep*. 2007;30(4):433–442. <https://doi.org/10.1093/sleep/30.4.433>

Circadian review

Coogan AN, McGowan NM. A systematic review of circadian function, chronotype and chronotherapy in attention deficit hyperactivity disorder. *ADHD Atten Def Hyp Disord*. 2017;9(3):129–147. <https://doi.org/10.1007/s12402-016-0214-5>

Tier Two: QbTest+ validation

Edebol H, Helldin L, Norlander T. Measuring adult attention deficit hyperactivity disorder using the Quantified Behavior Test Plus. *PsyCh J*. 2013;2(1):48–58.

Tier Two: QbTest clinical comparators

Brunkhorst-Kanaan N, et al. The Quantified Behavioral Test — a confirmatory test in the diagnostic process of adult ADHD? *Front Psychiatry*. 2020;11:216. <https://doi.org/10.3389/fpsy.2020.00216>

Tier Three: smartphone passive monitoring

Sankesara H, Denyer H, Sun S, et al. Identifying digital markers of attention-deficit/hyperactivity disorder (ADHD) in a remote monitoring setting: prospective observational study. *JMIR Form Res.* 2025;9:e54531. <https://doi.org/10.2196/54531>

Physiological cost and pain attribution

Hoffman KM, Trawalter S, Axt JR, Oliver MN. Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between Blacks and Whites. *Proc Natl Acad Sci USA.* 2016;113(16):4296–4301. | McEwen BS. Protective and damaging effects of stress mediators. *N Engl J Med.* 1998;338(3):171–179. | Geronimus AT, Hicken M, Keene D, Bound J. “Weathering” and age patterns of allostatic load scores among Blacks and Whites in the United States. *Am J Public Health.* 2006;96(5):826–833.

Strength and composure norms

Woods-Giscombé CL. Superwoman Schema: African American women’s views on stress, strength, and health. *Qual Health Res.* 2010;20(5):668–683.

Device equity

Colvonen AD. Limiting racial disparities and bias for wearable devices in health science research. *Sleep.* 2020;43(10):zsaa159.

Regulatory: EU MDR

Regulation (EU) 2017/745 on medical devices. Official Journal of the European Union, 5 April 2017. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32017R0745>

Regulatory: EMA qualification

European Medicines Agency. Qualification of novel methodologies for drug development: guidance to applicants. EMA/CHMP/SAWP/72894/2008. <https://www.ema.europa.eu/en/qualification-novel-methodologies-drug-development>

Regulatory: ICH E6(R3)

ICH E6(R3) Guideline for Good Clinical Practice. International Council for Harmonisation, 2025. Principles and Annex 1 effective in the EU 23 July 2025 (EMA). <https://www.ema.europa.eu/en/ich-e6-good-clinical-practice-scientific-guideline>

Regulatory: GDPR and BDSG

Regulation (EU) 2016/679 (GDPR), Articles 9 and 89. <https://gdpr-info.eu/art-9-gdpr/> | Bundesdatenschutzgesetz (BDSG), sections 22 and 27.